Response to Reviewers

Training-Free Ranking from Pairwise Comparisons via Acyclic Graph Construction

Journal of Supercomputing — Major Revision

Soroush Vahidi

Cover note to the Editor

Dear Editor,

Thank you for arranging this careful review. We have prepared a major revision addressing all comments from Reviewers 1–4. The principal changes are:

- title revised to remove the word "Scalable";
- corrected add-back implementation using exact reachability-aware cycle safety (replacing the submitted fixed-topological proxy);
- secondary weighted min-cut structural exchange with proofs and regime-specific empirics;
- rebuilt baseline comparisons from a single canonical source and replaced headline tables with OURS-Reach (resolving the submitted Table 4/5 inconsistency and the topo-proxy/add-back mismatch);
- direct classical runtime comparisons (showing that OURS is slower than lightweight classical methods);
- comprehensive structural ablation, parameter sensitivity, timeout/coverage analysis, and family-aware statistics;
- clarified ranking–MWFAS theory and DF03 guarantee boundaries;
- manuscript-wide rewriting with a dedicated non-repetition pass.

No new scientific experiments remain open. The point-by-point responses below cite the revised manuscript sections, propositions, tables, and figures.

Sincerely,
Soroush Vahidi

General reply to all reviewers

Dear Editor and Reviewers,

Thank you for the careful and constructive reviews. We agree that the original presentation overstated novelty relative to prior local-ratio/add-back lineage, left the Demetrescu–Finocchi guarantee ambiguous under practical timeouts, and did not provide the ablation, classical runtime, and statistical evidence needed for a major revision. We have substantially revised the manuscript accordingly. Throughout, we distinguish *prior* algorithmic lineage, an *implementation-fidelity* correction, *new theory*, a *secondary* min-cut repair, and an *expanded* empirical study.

Reviewer 1

Comment 1 (Positioning / novelty). *The contribution relative to prior feedback-arc-set / ranking work was not clearly delineated; the paper appeared to present the pipeline as more novel than supported.*

Response. We agree that the original positioning was insufficient. We completely rewrote the Introduction, restructured Related Work (including an explicit relation to Vahidi & Koutis, arXiv:2412.16181), and added a compact novelty-separation table. We no longer present local-ratio cycle breaking, exact cycle-safe reinsertion, weight-ordered add-back, or ternary refinement as newly invented. The title no longer contains "Scalable."

Changes in the manuscript: Sections 1 and 1.2; Table 1; title page.

Comment 2 (Ablation / sensitivity). *Provide ablations of Phase 1 only, Phase 1 with reinsertion, and the full pipeline; report upset metrics and runtime; study scale/density and sensitivity to zero tolerance, reinsertion passes, and refinement iterations.*

Response. We added a structural ablation with human-readable stage labels: A0 = Phase A only; A1 = Phase A + submitted topological-proxy add-back; A2 = Phase A + exact reachability add-back; A4 = A2 + refinement; A5/A6 = optional min-cut on top of A2/A4. The principal empirical method is OURS-Reach = A4 (reachability + refinement; min-cut off). On non-Finance common completions, exact reachability add-back (A2) improves the ablation `upset_simple` versus Phase A only (A0) with W/T/L 76/0/1 ($n = 77$; Holm $p = 2.4\times10^{-12}$), and versus the submitted topological proxy (A1) with 32/0/1 ($n = 33$). Legacy multipass topo-proxy passes P2/P3 add little beyond P1; `zero_tol` is stable; refinement saturates quickly (R1≈R2). Non-Finance A4 runtimes are roughly 0.01–1.2 s (median ≈ 0.57 s) for $n \leq 602$; Finance is the large-dense boundary (Section 3.5).

Changes in the manuscript: Sections 3.1, 3.5–3.8; Table 7; Figures 1–2.

Comment 3 (Algorithm readability). *The algorithm description needs complete, readable pseudocode and explicit parameters (ties, tolerances, timeouts).*

Response. We inserted a unified Algorithm 1 covering deterministic cycle search, residual updates, zero tolerance, forced-progress safeguards, wall-clock checks, exact reachability reinsertion, optional min-cut, refinement, and identity fallback, together with a consolidated parameter table.

Changes in the manuscript: Sections 2.3 and 2.8; Algorithm 1; Table 2.

Comment 4 (Theory / complexity / guarantees). *Clarify the ranking–MWFAS relationship, complexity claims, and whether the practical method inherits the Demetrescu–Finocchi approximation guarantee; provide an error discussion for fallback ordering.*

Response. Proposition 1 proves optimum-value equivalence $\mathrm{OPT}_{\mathrm{rank}} = \mathrm{OPT}_{\mathrm{MWFAS}}$ (many-to-many, not a bijection). For the audited Phase A implementation, the idealized full-completion complexity is $O(mn + m^2)$. Section 2.9 separates the idealized local-ratio guarantee from the practical finite-budget pipeline: we do *not* claim the DF03 guarantee for prematurely terminated runs. Proposition 4 shows that identity-order fallback admits no nontrivial general multiplicative bound.

Changes in the manuscript: Sections 2.2, 2.9–2.11; Propositions 1 and 4; Remark on DF03.

Comment 5 (Applications / future work). *Provide more concrete future directions and application context.*

Response. We added dedicated Limitations and Future Work sections covering denser-graph theory, SCC/parallel engineering, candidate selection for min-cut, and application settings (preference aggregation, multi-source sensors, industrial ranking) as directions, not evaluated domains.

Changes in the manuscript: Sections 4–6.

Reviewer 2

Comment 1 (Novelty). *Clarify what is scientifically new relative to prior MWFAS / ranking work.*

Response. As in our reply to Reviewer 1 (Comment 1), we now separate prior lineage, fidelity correction, new theory, secondary min-cut, and expanded empirics (Table 1; Sections 1–2).

Comment 2 (Add-back contribution). *The submitted add-back / INS variants do not appear to contribute clearly; please diagnose and strengthen this stage.*

Response. We agree that the submitted fixed-topological proxy was too conservative and that multipass INS2/INS3 were not major quality innovations. Exact reachability restores the intended cycle-safety test and materially improves rankings in the ablation (A0→A2: 76/0/1; A1→A2: 32/0/1). The principal Tables 4–6 now report OURS-Reach (exact reachability + refinement), not the submitted topo-proxy. INS multipass is retained only historically. As a secondary mechanism we added a weighted min-cut exchange with monotone weighted-FAS improvement (Proposition 3; Section 3.9); min-cut is not part of the headline method.

Changes in the manuscript: Sections 2.5–2.6, 3.7–3.9; Algorithms 1–2; Table 7.

Comment 3 (GNN protocol / timing fairness). *Clarify what GNN runtimes include and whether hardware is comparable.*

Response. Archived GNN timings measure end-to-end training+evaluation, not inference-only latency. The exact archived GPU model ledger is unavailable; the manuscript therefore qualifies hardware comparability and does not claim matched stage-by-stage inference fairness.

Changes in the manuscript: Section 3.1; runtime discussion in Section 3.4.

Comment 4 (Timeout bias / missingness). *Timeouts and missing values may bias aggregates; please report coverage honestly.*

Response. Pairwise common-completion comparisons are now primary. Coverage is reported explicitly on the 78 loadable suite members (e.g., OURS 77/78; classical typically 78/78; DIGRAC/ib 61/78). Finance is separated rather than averaged into quality macros. We do not impute NaNs as scores.

Changes in the manuscript: Sections 3.1, 3.4, 3.5.

Comment 5 (Oracle best-in-suite). *Best-in-suite / oracle envelopes should not be treated as deployable competitors.*

Response. We agree. Direct fixed-method comparisons are primary; oracle envelopes are de-emphasized and labeled as non-deployable post-hoc context only.

Changes in the manuscript: Section 3.1; captions of Tables 4–5.

Comment 6 (Statistics / dataset dependence). *Provide formal paired statistics and address Basketball-dominated dependence.*

Response. We report W/T/L, median paired deltas, and—where exported—Wilcoxon tests with Holm adjustment and bootstrap CIs. Family-aware equal-family macros, leave-one-family-out, and Basketball-collapsed analyses are included. SpringRank comparisons are acknowledged as LOFO-fragile; BTL's `upset_ratio` advantage persists across families.

Changes in the manuscript: Sections 3.2–3.3; Tables 4–5, 7.

Comment 7 (Deterministic repetitions). *Ten deterministic repetitions should not be interpreted as ranking robustness.*

Response. We agree. For deterministic non-GNN methods, repeats characterize runtime variability and timeout repeatability, not independent ranking observations.

Changes in the manuscript: Sections 3.1 and 4.

Comment 8 (Scalability). *The word "scalable" is not supported without qualification given dense/large cases.*

Response. We removed "Scalable" from the title and qualify empirical scope to the evaluated sparse and moderately dense graphs in our suite, with Finance as an explicit boundary.

Changes in the manuscript: Title; Sections 3.5–3.6, 4.

Comment 9 (Presentation / repetition). *The manuscript is repetitive and hard to navigate.*

Response. We consolidated the experimental protocol, separated Limitations from Conclusion, and performed a manuscript-wide repetition audit (novelty, DF03, INS, Finance, runtime accounting, contributions).

Changes in the manuscript: Throughout; Sections 3.1, 4–5.

Reviewer 3

Comment 1 (Ineffective add-back / INS1–3). *OURS-MFAS and INS variants appear nearly identical; add-back does not seem to help.*

Response. We agree with this diagnosis of the submitted manuscript: archived OURS-MFAS and INS3 were effectively duplicate topo-proxy configurations, and the fixed-topological rule was too conservative—so the submitted flagship add-back yielded no measurable ranking benefit. The revised Phase B uses exact reachability-aware reinsertion, and the principal comparison tables now evaluate OURS-REACH (reachability + refinement) under the same GNNRank upset metrics as the baselines. Ablations show A2 improves the stage metric over Phase A on 76/77 datasets and over the legacy proxy on 32/33. Multipass INS2/INS3 are historical only; optional min-cut remains secondary.

Changes in the manuscript: Sections 2.5, 3.1, 3.7–3.8; Table 7.

Comment 2 (Table 4 / Table 5 inconsistency). *Submitted baseline tables were inconsistent (including classical values that did not match).*

Response. We agree and take responsibility for this error. Stale aggregation/pivot logic produced inconsistent submitted summaries. The revision rebuilds all baseline tables from one canonical per-method leaderboard export under pairwise common-completion rules. For example, the canonical SpringRank median `upset_simple` in the verified full-suite export is 0.802724. Submitted unpaired Table 4/5 aggregates are not reused.

Changes in the manuscript: Section 3.1 (canonical source statement); Tables 4–5.

Comment 3 (Runtime versus classical methods). *Compare runtime directly against classical baselines, not only against GNNs.*

Response. We now report paired runtime W/T/L for OURS-Reach against classical methods. It is slower than most lightweight classical estimators (e.g., median ratios $\approx 5.7\times$ vs SpringRank and $\approx 170\times$ vs PageRank) and comparable to SyncRank (slightly faster on the median paired ratio). The runtime advantage is relative to archived trained GNN end-to-end runs (about 37–45$\times$ faster on 60/60 common completions vs DIGRAC/ib), with the protocol qualification noted above.

Changes in the manuscript: Section 3.4; Table 6.

Comment 4 (Accuracy–runtime tradeoff / overclaiming). *Claims of accuracy and runtime advantage need to be metric-specific and honest.*

Response. We revised claims to be metric-specific for OURS-Reach: strong simple/naive upset performance against the listed baselines; SpringRank remains family-sensitive despite a significant pairwise Holm test on `upset_simple`; BTL remains stronger on `upset_ratio`. Ablations document why the submitted topo-proxy add-back failed and how exact reachability contributes. We do not claim universal dominance. Together with Comments 1–3, the three concrete requests—reconcile tables, adopt reachability-aware insertion in the flagship evaluation, and add classical runtime comparisons—are implemented.

Changes in the manuscript: Abstract; Sections 3.2–3.4, 5; Tables 4–6.

Reviewer 4

Comment 1 (Scientific novelty). *What exactly is scientifically new?*

Response. The Introduction and Related Work now answer this directly via prior/new separation and Table 1: formal ranking–MWFAS clarification and guarantee boundaries; fidelity restoration of exact add-back; secondary min-cut exchange; expanded empirical evaluation. Prior local-ratio/add-back/refinement components are credited, not reinvented.

Changes in the manuscript: Sections 1–2; Table 1.

Comment 2 (DF03 approximation guarantee). *Does the practical implementation inherit the Demetrescu–Finocchi guarantee?*

Response. We clarify that the published guarantee applies to the idealized local-ratio procedure under its completion assumptions. The practical pipeline adds wall-clock termination, floating-point tolerances, forced-progress safeguards, and fallback output; therefore prematurely terminated practical runs do *not* automatically inherit the DF03 guarantee (Remark in Section 2.9). Proposition 4 addresses fallback error.

Changes in the manuscript: Sections 2.9–2.10, 4.

Comment 3 (Multipass INS). *INS1/INS2/INS3 appear oversold.*

Response. Sensitivity analysis shows P1 restores many edges while P2/P3 add little ranking quality. Multipass INS variants are no longer framed as a core contribution.

Changes in the manuscript: Sections 2 (opening), 3.1, 3.8.

Comment 4 (GNN protocol). *Clarify GNN timing/accounting.*

Response. As in our reply to Reviewer 2 (Comment 3): end-to-end trained-GNN protocol, not inference-only; hardware ledger unavailable; claims qualified in Section 3.1.

Comment 5 ("Scalable"). *The title/text overstate scalability.*

Response. We removed "Scalable" from the title and describe empirical behavior on the evaluated sparse and moderately dense graphs in our suite, with Finance as an explicit large-dense boundary.

Changes in the manuscript: Title; Sections 3.5–3.6, 4.

Comment 6 (Writing / repetition). *The writing is repetitive and sentences are too long.*

Response. We rewrote the Introduction, consolidated protocol and limitations, removed duplicated novelty/DF03/INS/Finance explanations, and shortened long passages while preserving technical precision.

Changes in the manuscript: Throughout; Sections 1, 3.1, 4–5.

Comment 7 (MWFAS backbone / cycle-selection ablations). *Ablate the MWFAS backbone and related algorithmic choices.*

Response. The A0–A6 structural ablation addresses backbone stages (Phase A only; topo-proxy add-back; exact reachability; refinement; optional min-cut). Cycle-selection sensitivity (C0/C1) shows a small but detectable effect on the A4 path ($|\mathrm{med}\,\Delta| \approx 10^{-3}$); we report it without overstatement.

Changes in the manuscript: Sections 3.7–3.8; Table 7; Figure 2.

Closing

We thank the reviewers again for comments that substantially improved the clarity, correctness, and evidence standards of the manuscript. We hope the revised version addresses all concerns.